

JUMMA MOHAMMAD TELI

Machine Learning Engineer | Python · MLOps · Model Serving · MLflow · Docker · Kubernetes · Airflow · FastAPI · CI/CD · AWS

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PROFESSIONAL SUMMARY

Machine Learning Engineer with an MSc in Computer Science and 7+ years building data-driven, production-grade analytical systems. I take models from notebook to production: feature engineering, hyperparameter tuning, packaging, serving, and monitoring. Author of a 10-project end-to-end MLOps portfolio (MLflow, DVC, Optuna, FastAPI, Docker, Airflow, Kubernetes) with 132 passing tests on real-world datasets, plus deployed ML services for readmission risk, churn, and energy-demand forecasting. Strong software-engineering discipline — Python packaging, pytest, type checking, linting, and GitHub Actions CI/CD — combined with deep SQL and data-pipeline experience across financial services, telecom, and public-sector environments. Comfortable owning the full ML lifecycle: data ingestion, training, evaluation (SHAP, drift detection), reproducible pipelines, REST APIs, containerisation, and orchestration.

TECHNICAL SKILLS

- **Languages:** Python, SQL (T-SQL), Bash, PySpark
- **ML Libraries:** Pandas, NumPy, Matplotlib, sentence-transformers
- **Serving:** FastAPI, Flask, Streamlit, REST APIs, Uvicorn
- **Containers:** Docker, Kubernetes, Helm, docker-compose
- **Monitoring:** Drift detection (PSI / KS-test), Prometheus, A/B testing
- **Data Eng:** ETL/ELT, star/snowflake schemas, data warehousing
- **NLP:** Embeddings, UMAP, HDBSCAN, clustering, sentiment analysis
- **Version Control:** Git, GitHub, branching, code review, PRs
- **Big Data:** Spark/PySpark, Databricks, partitioning, parallel jobs
- **Linux / CLI:** Linux, shell scripting, cron, SSH, Makefiles
- **ML / DL:** Scikit-learn, XGBoost, PyTorch (basics), SHAP
- **MLOps:** MLflow, DVC, Optuna, Feature Stores, Model Registry
- **Orchestration:** Airflow, scheduled retraining pipelines
- **CI/CD:** GitHub Actions, pytest, ruff, mypy, quality gates
- **Cloud:** AWS (S3, Redshift, Lambda), GCP, Databricks, Render
- **Databases:** Teradata, SQL Server, PostgreSQL, MySQL, SSMS
- **Stats:** Regression, forecasting, A/B testing, feature selection
- **BI & Viz:** Power BI (DAX, Power Query), Tableau, Cognos, SSRS
- **Environments:** conda, venv, pip, YAML config, .env management
- **Practices:** Agile/Scrum, JIRA, TDD, reproducibility, documentation

KEY ML & ENGINEERING PROJECTS

End-to-End MLOps Portfolio — 10 Projects · MLflow · FastAPI · Docker · Airflow · Kubernetes · DVC · Optuna

Built a 10-project production MLOps portfolio covering the full ML lifecycle: multi-model tournament with experiment tracking, scheduled retraining (DVC), feature engineering, hyperparameter tuning (Optuna), FastAPI + Docker model serving, a feature store, drift detection (PSI/KS-test), A/B testing, Airflow orchestration, and a Kubernetes serving platform with Helm. 132 passing tests across all projects on real UCI datasets (Bank Marketing 41K rows, Online Retail II 1M+ rows).

GitHub: github.com/jumma786 (mlops-model-tournament, mlops-model-serving, mlops-k8s-platform + 7 more)

Hospital Readmission Risk — Model + Live API · XGBoost · SHAP · FastAPI · Render · T-SQL

Trained an XGBoost readmission model achieving AUC 0.907 with SHAP explainability on synthetic EHR data, backed by a T-SQL star-schema pipeline. Packaged and deployed the model as a live readmission-risk prediction REST API via FastAPI on Render.

GitHub: [hospital-readmission-api](#) · [hospital-analytics-end-to-end](#)

UK Energy Demand Forecasting — ML Service · XGBoost · FastAPI · Streamlit · GitHub Actions

End-to-end ML pipeline forecasting UK electricity demand with XGBoost, exposed through a FastAPI prediction service and a live Streamlit dashboard. Fully tested codebase with CI, demonstrating reproducible training and deployment.

GitHub: [uk-energy-forecasting](#)

UK Job Market NLP & Salary Modelling · sentence-transformers · UMAP · HDBSCAN · XGBoost

Embedded 3,653 UK job postings with sentence-transformers and clustered them into 17 role archetypes via UMAP/HDBSCAN. Built an XGBoost salary model ($R^2=0.51$) showing real salaries are more predictable than platform-predicted ones.

GitHub: [uk-data-analyst-job-market-nlp](#)

Retail Churn Prediction at Scale · Python · XGBoost · SQL Server · leakage-safe features

Built an XGBoost churn model on 1.07M UK retail transactions using leakage-safe, time-based features, identifying £1.26M revenue exposure across the top 100 at-risk customers. Included SQL Server RFM segmentation and a 4-page reporting layer.

GitHub: [online-retail-analytics](#)

PySpark Sales Analysis — Productionised Package · PySpark · Databricks · pytest · ruff · mypy · CI/CD

Refactored a Databricks notebook into a production Python package (11 modules) with a 24-test pytest suite, GitHub Actions CI, and a full quality gate (ruff, mypy) — demonstrating software-engineering rigour applied to data workloads.

GitHub: [pyspark-sales-analysis](#)

Portfolio Risk Analyser — Live Web App · Python · Flask · yfinance · Render

Built and deployed a portfolio risk and concentration analyser computing sector exposure, volatility, correlation matrices, and maximum drawdown, served as a live Flask web app on Render.

GitHub: [portfolio-risk-analyser](#) · [portfolio-risk-web](#)

Airline Operations Analytics — ETL & Star Schema · Python · SQL Server · Power BI · Data Quality

Ran a data-quality audit and rebuild on 495K BTS flight records: Python ETL into a SQL Server star schema with validation checks, surfacing DOT-standard metrics (on-time performance, delay causes, route efficiency) for downstream reporting.

GitHub: [airline-operations-analytics](#)

Investment Fund Industry Analysis · Python · SQL Server · Power BI

End-to-end pipeline on U.S. investment-fund trends (2010–2026): Python data processing, SQL Server modelling, and dashboards visualising fund performance, AUM distribution, and sector allocation.

GitHub: [investment-company-analysis](#)

PROFESSIONAL EXPERIENCE

Data Analyst (Contract, Remote) | UBS Oct 2024 – Present

- Automate recurring data-extraction and reporting workflows in Python and SQL across large financial, operational, and risk datasets, saving ~8 hours/week and improving accuracy and consistency.
- Develop and optimise 50+ complex SQL queries, views, and analytical datasets across Teradata and SQL Server, feeding regulatory reporting, KPI measurement, and forecasting models across business functions.
- Perform data profiling, cleansing, and validation on multi-million-row tables extracted from the enterprise Teradata data warehouse, building reproducible pipelines that ensure data integrity for downstream analytics and modelling.
- Collaborate with risk, compliance, and finance teams to translate business requirements into technical specifications and analytical/data solutions.
- Maintain data documentation, dictionaries, and lineage maps to support governance, reproducibility, and knowledge transfer.
- Build reusable, parameterised Python scripts and modular SQL templates that standardise data preparation steps across multiple reporting workflows.
- Apply version control (Git) and peer review to analytical code, improving maintainability and reducing errors in recurring deliverables.

Data Analyst | IT Skills, London, UK Oct 2023 – Oct 2024

- Automated ETL workflows in Talend Open Studio and Python, reducing manual processing time by 40% and improving delivery timeliness by 20%.
- Wrote and optimised complex SQL for validation, cleansing, and extraction across large datasets, achieving 99%+ accuracy for downstream reporting and analysis.
- Integrated multiple online and offline sources into a unified data warehouse, improving reliability and cross-team data accessibility.
- Built interactive Power BI dashboards visualising KPIs across departments, increasing reporting efficiency by 30%.
- Developed Python scripts for data cleansing, transformation, and feature preparation, replacing manual spreadsheet steps with repeatable code.
- Containerised a recurring reporting script with a simple config file, making it portable across environments without manual setup.
- Wrote unit tests for core data-transformation functions, catching regressions before they reached production reports.
- Created scheduled reports and automated refreshes, streamlining distribution to 50+ end users and reducing ad-hoc reporting requests.

Data Analyst | Skyway Internet Services, Nagaur, India Apr 2019 – Jul 2023

- Built predictive models to forecast customer churn and key business metrics, enabling evidence-based resource allocation and strategic planning.
- Designed and maintained SQL-based data pipelines processing 100K+ daily records, ensuring timely data availability for reporting and modelling.
- Performed exploratory data analysis and customer segmentation (RFM) to identify high-value accounts and inform targeted retention campaigns.
- Implemented automated data validation and quality checks, rapidly resolving discrepancies to strengthen dataset reliability.
- Engineered model features from raw operational logs and evaluated forecasts against actuals to iteratively improve predictive accuracy.
- Retrained and re-evaluated forecasting models on a monthly cadence as new data arrived, tracking performance drift over time.
- Packaged model scoring logic into reusable SQL/Python scripts so non-technical staff could refresh predictions without engineering support.
- Built Power BI and Tableau dashboards to communicate model outputs and KPIs to non-technical stakeholders, improving data literacy.

IT Technician | BSF (Border Security Force), Delhi, India Apr 2016 – Jan 2019

- Operated and maintained secure communication systems and implemented data backup/security procedures across multiple units.
- Provided IT support and trained 30+ staff on new systems and protocols, improving operational efficiency.
- Managed inventory and equipment records in structured databases and spreadsheets, ensuring accurate, auditable asset tracking.

EDUCATION

MSc Internet of Things and Future Networks — Merit
Queen Mary University of London, UK
Sep 2023 – Sep 2024

BSc Information Technology
Sabarmati University, Gujarat, India
Sep 2012 – Jun 2015

CERTIFICATIONS

- Introduction to Data Analytics — IBM
- Tools for Data Science — IBM

- [Excel Basics for Data Analysis — IBM](#)
- [Introduction to Artificial Intelligence \(AI\) — IBM](#)
- [Power BI for Finance — LinkedIn Learning](#)

- [Data Visualization and Dashboards with Excel and Cognos — IBM](#)
- [SQL \(Basic\) — HackerRank](#)
- [Data Science Methodologies: Making Business Sense — LinkedIn Learning](#)